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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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EXAMINER
BRAINARD, TIMOTHY A

ART UNIT	PAPER NUMBER
3648	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docketing@slatermatsil.com

Office Action Summary	Application No. 17/396,032	Applicant(s) Servadei et al.	
	Examiner TIMOTHY A BRAINARD	Art Unit 3648	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 3/11/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-25 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-4,6-11,13 and 17-25 is/are rejected.

8) ☑ Claim(s) 5,12 and 14-16 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 8/6/2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 102

2. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

3. Claim(s) 1-2, 7, 9, 17-18, 21-24 is/are rejected under 35 U.S.C. 102(a)(2) as being anticipated by Chen et al (US 20220196798).

4. Regarding claim 1, **Chen** teaches a method a method comprising: receiving first radar data from a millimeter-wave radar sensor (abs, “generate a dataset representative of the scene and a radar detector configured to determine an object parameter of the object using a machine learning algorithm and the dataset); receiving a set of hyperparameters with a radar processing chain (para 188, “The machine learning algorithm may include the vector of weighted values.”); generating a first radar processing output using the radar processing chain based on the first radar data and the set of hyperparameters (para 195, “The radar detector 1110 may determine object parameters of the object in the scene. The radar detector 1110 may determine the object parameters using a machine learning algorithm and the dataset. The radar detector 1110 may detect the object in the scene based on the determined object parameters.”), updating the set of hyperparameters based on the first radar processing output using a hyperparameter selection neural network (para 200, “The radar detector 1110 may use the machine learning algorithm including the adjusted weighted values and the dataset to determine the another set of object parameters”), receiving second radar data from the millimeter-wave radar sensor (para 200,

"The radar detector 1110 may determine another set of object parameters of the object in the scene"); and generating a second radar processing output using the radar processing chain based on the second radar data and the updated set of hyperparameters (para 200, "The radar detector 1110 may determine another set of object parameters of the object in the scene").

5. Regarding claim 2, **Chen** teaches first radar processing output comprises a first radar image (para 361, "According to various embodiments, an end-to-end learnable architecture is provided that reconstructs (e.g. 4D) radar images from (raw) digital (radar) reception data received from the radar ADC").

6. Regarding claim 7, **Chen** teaches the radar processing chain comprises a tracker, wherein the set of hyperparameters comprises a gating threshold hyperparameter, and wherein receiving the set of hyperparameters comprises receiving the gating threshold hyperparameter with the tracker (para 371, "a detector compares the output of the AOA determination, e.g. Doppler FFT output values for each bin combination (e.g. for each range-Doppler bin) with a threshold and discards those bin combinations where the value if below the threshold" and "pipeline of perception may then be used to further perform tasks such as object detection classification, segmentation and tracking on the point cloud.")).

7. Regarding claim 9, **Chen** teaches training the hyperparameter selection neural network (para 387, "for training the neural networks DeepRD and DeepAOA, they can be first trained separately with independently collected or generated training samples in respective Range-Doppler and angular dimensions").

8. Regarding claim 17, **Chen** teaches the first radar processing output comprises a first radar image (para 404, " the two reconstruction stages (RD reconstruction, implemented by the CRD and the DeepRD and the AOA reconstruction implemented by the CAOA and the DeepAOA) of the baseband processing and image reconstruction pipeline 2900 in sequence can yield the overall 3D/4D point cloud or voxel representation for a given scene for a given frame"); the radar processing chain comprises a tracker (para 404, "a 5D pipeline to process the

time-series data of the sequence for tracking targeted objects in a scene, in an end-to-end fashion"), the method further comprises tracking a target using the tracker based on the first radar image and the set of hyperparameters; and updating the set of hyperparameters based on the first radar processing output using the hyperparameter selection neural network comprises updating the set of hyperparameters based on the first radar image using [[a]] the hyperparameter selection neural network (para 564, " the radar detector is further configured to adjust one or more adjusted weighted values of the plurality of adjusted weighted values based on a difference between the second error value and the first error value to reduce the error value." and para 565, "the dataset includes a complex values image.")

9. Regarding claim 18, **Chen** teaches the target is a human target (para 181, "The radar device may determine (e.g., estimate) object parameters of the objects, such as locations, relative speeds and even classify the type of objects (vehicle, pedestrian etc.)")

10. Regarding claim 21, **Chen** teaches millimeter-wave radar system comprising: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals (para 272), and generate radar data based on the reflected radar signals, wherein the radar data comprises first radar data and second radar data; and a processing system comprising: a receiver configured to: receive the first radar data from the millimeter-wave radar sensor, receive the second radar data from the millimeter-wave radar sensor, a radar processing chain configured to: generate a first radar processing output based on the radar data and a set of hyperparameters, and a hyperparameter selection neural network configured to: update the set of hyperparameters based on the first radar processing output, wherein the radar processing chain is configured to generate a second radar processing output based on the second radar data and the updated set of hyperparameters (para 188-200)

11. Regarding claim 22, **Chen** teaches the radar processing chain is configured to track a target based on the radar data (para 186)

12. Regarding claim 23, **Chen** teaches the radar data comprises a plurality of consecutive frames, and wherein the hyperparameter selection neural network is configured to update the set of hyperparameters every n frames of the radar data, wherein n is a positive integer greater than or equal to 1 (para 200, "the radar detector may iteratively perform the operations discussed above to reduce the error value").

13. Regarding claim 24, **Chen** teaches the hyperparameter selection neural network comprises an initial layer, a plurality of convolutional layers, and a plurality of dense layers (para 323, "The first neural network 2001 includes a first sequence of neural network layers 2004, 2005. For example, the first neural network 2001 is a convolutional network and (at least some of) the neural network layers 2004, 2005 are convolutional layers" and para 330 "For each segment of the segmentation 2007, the part of the feature maps 2006 corresponding to the 4D-bins of the segment as well as the 4D-bins of the segment itself are fed to the second neural network 2002 which contains a second sequence of neural network layers 200").

Claim Rejections - 35 USC § 103

14. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

15. Claim(s) 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 1 above, and further in view of **Beaudoin** et al (US 20220067884).

16. Regarding claim 3, **Beaudoin** teaches generating the first radar image using the radar processing chain based on an aggregation of a last m frames of the first radar data, wherein m is a positive integer greater than 1 and lower than or equal to 32 (para 12, "step of filtering the at least one unfiltered image comprises one of transforming the at least one unfiltered image in a

colorspace and adding frames together"). It would have been obvious to modify **Chen** to include generating the first radar image using the radar processing chain based on an aggregation of a last m frames of the first radar data, wherein m is a positive integer greater than 1 and lower than or equal to 32 because it a substitution of the well-known method to develop radar images of **Beaudoin** with the well-known method of making radar images of **Chen** with no new or unexpected results.

17. Claim(s) 4 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 2 above, and further in view of **Sanderovich** et al (US 10958287).

18. Regarding claim 4, **Sanderovich** teaches generating the first radar image by masking a range-angle image (RAI) or range-Doppler image (RDI) based on an output of an order statistics (OS) constant false alarm rate (CFAR) (OS-CFAR) detector (col 3, lines 47-53, "Each radar scan can produce radar data in several dimensions, including azimuth, elevation, time, Doppler, and range, for example. CFAR can be used to significantly compressed data by estimating a noise level within the radar data, then removing (compressing) values in the radar data"). It would have been obvious to modify **Chen** to include generating the first radar image by masking a range-angle image (RAI) or range-Doppler image (RDI) based on an output of an order statistics (OS) constant false alarm rate (CFAR) (OS-CFAR) detector because it a substitution of the well-known method to develop radar images of **Sanderovich** with the method of making radar images of **Chen** with no new or unexpected results.

19. Claim(s) 6 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 1 above, and further in view of **Cai** et al (US 11380108).

20. Regarding claim 6, **Cai** teaches generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap (col 28, lines 35-40, "the machine learning model can output a heat map associated with prediction probabilities based on image features. In some examples, at least one predicted trajectory can be determined based at least in part on the heat map"). It would have been obvious to modify

Chen to include generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap because it is applying a well-known tracking technique of Cai to a radar tracker of Chen to yield a predictable radar tracker.

21. Claim(s) 8 and 20 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 1 and 17 above, and further in view of **Ma** et al (US 20210365038).

22. Regarding claim 8 and 20, **Ma** teaches the tracker comprises an unscented Kalman filter (para 60, "Kalman filters, octrees, probabilistic multi-hypothesis tracking methods, and/or other suitable algorithms"). It would have been obvious to modify **Chen** to include the tracker comprises an unscented Kalman filter because it is applying a well-known tracking technique of Cai to a radar tracker of Chen to yield a predictable radar tracker.

23. Claim(s) 10-11 and 13 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 9 above, and further in view of **Smolyanskiy** et al (US 20220138568).

24. Regarding claim 10, **Smolyanskiy** teaches training the hyperparameter selection neural network comprises training the hyperparameter selection neural network using imitation learning (abs, "reinforcement learning is used to train at least one machine learning model (MLM) to control a vehicle by leveraging a deep neural network (DNN) trained on real-world data by using imitation learning to predict movements of one or more actors to define a world model"). It would have been obvious to modify **Chen** to include training the hyperparameter selection neural network comprises training the hyperparameter selection neural network using imitation learning because it is merely a applying a known neural network training technique of Smolyanskiy to the neural network of Chen to yield a predictable neural network.

25. Regarding claim 11, **Smolyanskiy** teaches training the hyperparameter selection neural network further comprises, after training the hyperparameter selection neural network using imitation learning, training the hyperparameter selection neural network using reinforcement learning (abs, "reinforcement learning is used to train at least one machine learning model (MLM) to control a vehicle by leveraging a deep neural network (DNN) trained on real-world

data by using imitation learning to predict movements of one or more actors to define a world model"). It would have been obvious to modify **Chen** to include training the hyperparameter selection neural network further comprises, after training the hyperparameter selection neural network using imitation learning, training the hyperparameter selection neural network using reinforcement learning because it is merely a applying a known neural network training technique of Smolyanskiy to the neural network of Chen to yield a predictable neural network.

26. Regarding claim 13, **Smolyanskiy** teaches training the hyperparameter selection neural network further comprises using curriculum learning during reinforcement learning (abs, "reinforcement learning is used to train at least one machine learning model (MLM) to control a vehicle by leveraging a deep neural network (DNN) trained on real-world data by using imitation learning to predict movements of one or more actors to define a world model"). It would have been obvious to modify **Chen** to include training the hyperparameter selection neural network further comprises using curriculum learning during reinforcement learning because it is merely a applying a known neural network training technique of Smolyanskiy to the neural network of Chen to yield a predictable neural network.

27. Claim(s) 19 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 17 above, and further in view of **Awtar** et al (US 20210114553).

28. Regarding claim 19, **Awtar** teaches the target is a heartbeat rate of a human (para 155, "the Bayesian classifier for a given set of inputs to the algorithm (i.e. passenger gender, age, height, weight, self-reported motion sickness susceptibility, physiological information such as heart rate and perspiration)"). It would have been obvious to modify **Chen** to include the target is a heartbeat rate of a human because it is merely a substitution of the object to be tracked of Chen with the object of Awtar to yield a predictable radar tracker.

29. Claim(s) 25 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Chen** as applied to claim 21 above, and further in view of **Paulsen** et al (US 20140050368).

30. Regarding claim 25, **Paulson** teaches the first radar processing output comprises a radar image and a masked radar image (para 157, " the edge 39 mask to radar image 141 is preferred may depend on the resolution of the specific radar image 141 to be use"). It would have been obvious to modify **Chen** to include the first radar processing output comprises a radar image and a masked radar image because it a substitution of the well-known method to develop radar images of **Paulson** with the well-known method of making radar images of **Chen** with no new or unexpected results.

Allowable Subject Matter

31. Claims 5, 12, 14-16 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

32. Any inquiry concerning this communication or earlier communications from the examiner should be directed to TIMOTHY A BRAINARD whose telephone number is (571)272-2132. The examiner can normally be reached Monday - Friday 7:00 a.m.-5 p.m.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Erin Heard can be reached on 571-272-3236. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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TIMOTHY A. BRAINARD
Primary Examiner
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33. /TIMOTHY A BRAINARD/
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